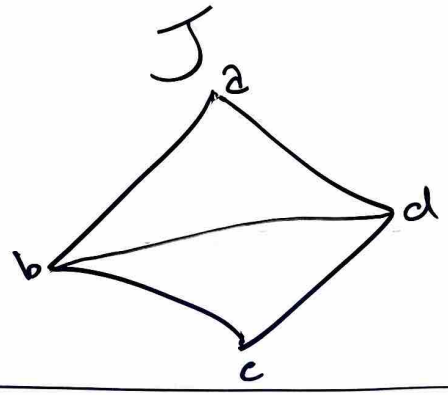
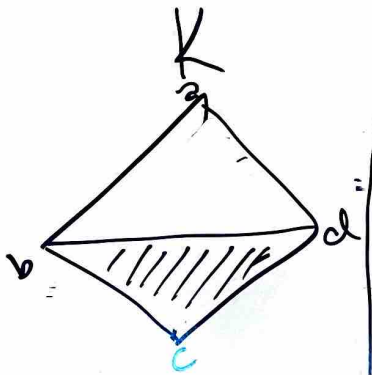
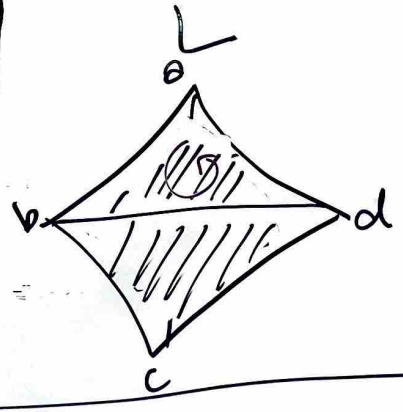
		
$C_0 = \{\alpha_0[a]+\alpha_1[b]+\alpha_2[c]+\alpha_3[d]$ s.t. $\alpha_i \in G\}$	$\mathbb{Z}_3 < a, b, c, d >$	$< a, b, c, d >$
C_1 $\{\alpha_0[a,b]+\alpha_1[a,d]+\alpha_2[b,d]+$ $+\alpha_3[b,c]+\alpha_4[c,d]$ s.t. $\alpha_i \in G\}$	$\mathbb{Z}_3 < [ab], [bc], [cd], [da], [bd] >$	$< [a,b], [b,c], [c,d], [d,a], [b,d] >$
C_2 $\{\emptyset\}$ (the trivial group)	$\mathbb{Z}_3 \langle [cbd] \rangle$	$< [a,b,d], [b,c,d] >$
C_3 $\{\emptyset\}$	$\emptyset$	$\emptyset$

	J	K	L
∂_0	see K	$C_0 \longrightarrow \{e\}$ $a \longmapsto e$ $b \longmapsto e$ $c \longmapsto e$ $d \longmapsto e$	see K
∂_1	$C_1 \to C_0$ $ab \mapsto a+2b = a-b$ $bc \mapsto b+2c = b-c$ $cd \mapsto c+2d = c-d$ $da \mapsto d+2a = d-a$ $bd \mapsto b+2c = b-c$ these are the same? Yes!	$C_1 \longrightarrow C_0$ $ab \longmapsto b+2a = b-a$ $bc \longmapsto c+2b = c-b$ $cd \longmapsto d+2c = d-c$ $da \longmapsto a+2d = a-d$ $bd \longmapsto d+2b = a-b$	$\partial_1 : C_1 \longrightarrow C_0$ $[a,b] \longmapsto b+2a$ $[b,c] \longmapsto c+2b$ $[c,d] \longmapsto d+2c$ $[d,a] \longmapsto a+2d$ $[b,d] \longmapsto d+2b$
∂_2	X	$C_2 \longrightarrow C_1$ $bcd \longmapsto cd+2bd+bc$	$\partial_2 : C_2 \longrightarrow C_1$ $[a,b,d] \longmapsto [b,d]+2[a,d]+[a,b]$ $[b,c,d] \longmapsto [c,d]+2[b,d]+[b,c]$

Why do only generators matter when defining Ω?

Suppose $x \in C_i$ is an arbitrary elt

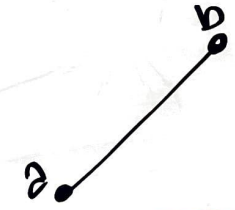
Then, $x = \sum_{\sigma_j \text{ generators}} \alpha_j \sigma_j$ by defn of gen. set

$$\Omega_i(x) = \sum_{\sigma_j} \Omega_i(\alpha_j \sigma_j)$$

$$= \sum_{\sigma_j} \alpha_j \boxed{\Omega_i(\sigma_j)}$$

(we know these

Example:



Coef Group G

$C_0(K;G) = C_0 = \langle a, b \rangle \cong G^2$, $C_0 = \langle a+b, a \rangle$ ($\langle\ \rangle$ means "generated by")

$C_1 = \langle [ab] \rangle \cong G$

$C_i = \{e\}$ $\forall i \geq 2$ (the trivial group)

$\partial_0 : C_0 \longrightarrow \{e\}$

$a \mapsto e$

$b \mapsto e$

$\partial_1 : C_1 \longrightarrow C_0$

$[ab] \longmapsto b - a$

$\partial_2 : C_2 \longrightarrow C_1$

$e \longmapsto e$

You can define maps by telling me where the generators go.

$\ker(\partial_0) = C_0 \cong G$ $\quad \ker(\partial_1) = \{e\}$ $\quad \ker(\partial_2) = C_2$

$\text{im}(\partial_0) = \{e\}$ $\quad \text{im}(\partial_1) = \{b-a\} \cong G$ $\quad \text{im}(\partial_2) = C_1$

$$\begin{cases} C_0 = \{\alpha_a a + \alpha_b b \mid \alpha_a, \alpha_b \in G\} \\ C_1 = \{\alpha[ab] \mid \alpha \in G_1\} \\ C_2 = \{e\} \end{cases}$$

with $\mathbb{Z}_3$ coef:

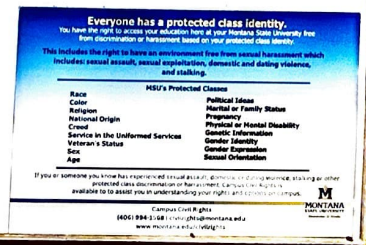
$$C_0 = \{\emptyset^{=e},\ a,\ b,\ a+b,\ a+2b,\ 2a+b,\ 2a+2b,\ 2a,\ 2b\}$$

$2(a+b)+(-1)a$
$=$
$a+2b$

means I am thinking of this as a set

$$X = 2a+b = 1(a+b)+1a$$

$$\partial(X) = \partial(2a+b)$$
$$= 2\partial(a)+\partial(b) = 2e+e = e$$




J
K
L
6
Ker ∂_2
im ∂_2
$\langle ab+bd+2da, bc+cd+2bd \rangle$
Ker ∂_1
$\langle bd+2ad+bd, cd+2bd+bc \rangle$
im ∂_1
Ker ∂_0
$\langle a, b, c, d \rangle$
im ∂_0
$\langle b+2a, c+2b, d+2c, a+2d, b+2d \rangle$
$\cong \langle b, c, d \rangle$